Deficiency:

Mineralocorticosteroids secreted by adrenal glands, regulate the metabolism of sodium.Decreased level of plasma sodium(hyponatremia) may be due to lack of aldosterone and loss from gastro-intestinal tract associated with conditions like diarrhoea.

9.2.4. Potassium

Potassium is the main intracellular cation, about 98% of potassium is present inside the cell.

Functions:

Many functions of potassium and sodium are carried out in co-ordination with each other.

1. Potassium maintains the intracellular osmotic pressure, water balance and acid-base balance.

2. Potassium, along with sodium influences neuromuscular activity of cardiac and skeletal muscles.

3. The glycolytic enzyme, pyruvate kinase requires potassium as cofactor.

4. It maintains the alkalinity of the bile and blood.

5. Protein synthesis is dependent on potassium levels inside the cells.

Sources:

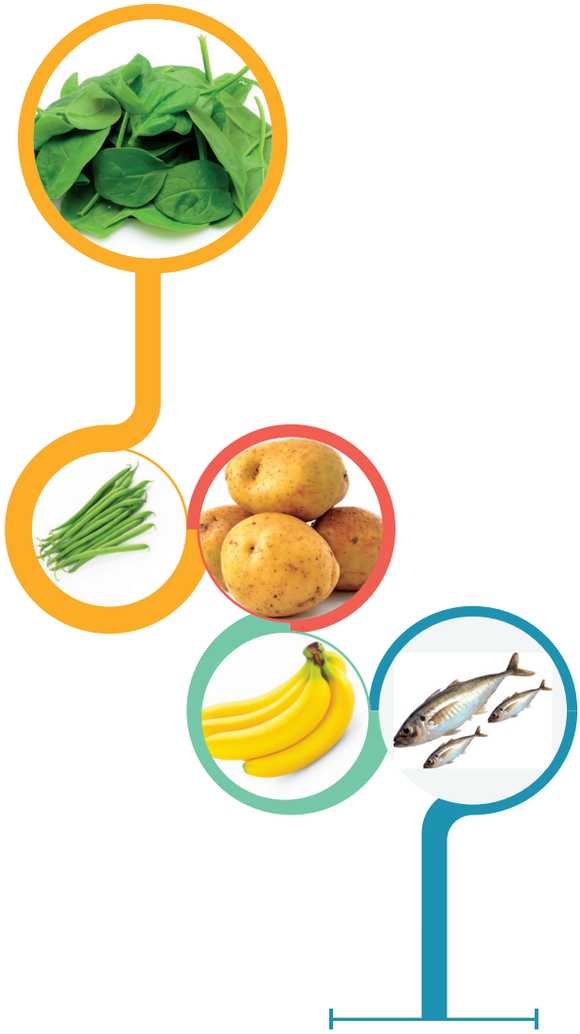


Figure 9.6 Sources of potassium

Animal sources of potassium include meat, fish, egg and milk. Vegetables like onion, carrot, fruits like apple, dates, banana, tender coconut water and grapes contain potassium.

Absorption and excretion:

Potassium is absorbed readily by passive diffusion from gastro intestinal tract and it is mainly excreted in the urine.

Unit 9.indd 235 21-05-2018 15:12:11